Adify Skills
n8n Automation Mastery

📘 CLASS NOTES

🔌 Module 3 · APIs, Webhooks & Integrations

DAY 09 of 25

Google Workspace & Messaging Automation

"Gmail, Sheets, Drive, Calendar + WhatsApp, Telegram, Slack."

⚡ AAJ HUM YEH BANAYENGE

Telegram Appointment Booking Bot

Customer Telegram par likhta hai "book 2026-12-05 16:00". Bot Google Calendar mein slot check karta hai — khaali ho to 30 min ki meeting book karke confirm karta hai, warna busy bata deta hai.

Google Sheets · Gmail · Drive · Calendar · Telegram · WhatsApp

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Indian businesses Google Workspace + WhatsApp/Telegram par chalte hain. Yeh class sabse zyada "turant paisa" wali skills deti hai.

🎯 Aaj ke baad aap yeh kar paoge

01 Google Sheets as a database: lookup, upsert

🌍 Attendance register: naam dhoondho (lookup), hai to tick, nahi to naya naam (upsert).
⚙️ Sheets → "Append or Update Row", match column = phone.

02 Gmail: read, label, auto-reply, attachments

🌍 Personal assistant jo inbox padhe, label lagaye, jawab de.
⚙️ Gmail Trigger (has:attachment) → label "Invoices" → reply.

03 Google Drive & Docs: file automation

🌍 Office almirah mein month-wise files.
⚙️ Drive: folder "Invoices/2026-12" create → file upload.

04 Google Calendar: booking automation

🌍 Doctor ki appointment diary.
⚙️ Calendar: Get availability → Create event (start, end, attendee).

05 Telegram bot from scratch

🌍 Apna chhota employee jo chat par jawab de.
⚙️ @BotFather → token → Telegram Trigger + Telegram Send.

06 Slack & WhatsApp Business (Cloud API) messaging

🌍 Office group (Slack) aur customer ka official WhatsApp.
⚙️ Slack #sales message · WhatsApp template "order_confirmed".

⏱️ Class ka plan (2 ghante)

10m	30m	50m	20m	10m

`0:00-0:10` **Recap**

🪄 Cricket match se pehle warm-up karte ho na? Recap dimaag ka warm-up hai.

`0:10-0:40` **Concept: Workspace APIs**

Scopes & quotas

🪄 Socho tumhare paas ek robot hai jo tumhari diary (Calendar), register (Sheet), aur phone (Telegram) sab sambhaal leta hai.

`0:40-1:30` **Live Build**

Appointment booking bot: Telegram → Calendar → Sheet → reminder

🪄 Aaj ki kahani yaad hai — 'Clinic ka smart receptionist bot'? Ab wahi kahani n8n mein banayenge. Har scene ek node hai: ek-ek block jodo aur har block ke baad dekho sab chal raha hai ya nahi.

`1:30-1:50` **Challenge**

Gmail attachment → Drive auto-sorter

🪄 Practice match — jitna yahan girenge, utna asli match mein kam girenge.

`1:50-2:00` **Q&A + Homework**

🪄 Ghar jaane se pehle bag check — sab samajh aaya? Homework diary mein likha?

Concepts — aasaan bhasha mein

🧠 Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Sheets as database

Lookup, append, update, upsert (row match karke update ya naya).

🪄 **Analogy:** Register book

CONCEPT 02

Gmail automation

Read, label, reply, attachments download.

🪄 **Analogy:** Personal assistant jo inbox sambhaale

CONCEPT 03

Telegram Bot

@BotFather se token, phir n8n se messages/buttons.

🪄 **Analogy:** Apna khud ka chatbot employee

CONCEPT 04

WhatsApp Cloud API

Meta Business se — template messages, 24-hr window rule.

🪄 **Analogy:** Official business WhatsApp

🔑 Aaj ka main idea

Google Workspace (Sheets, Gmail, Drive, Calendar) aur messaging apps (Telegram, WhatsApp, Slack) — yahi Indian businesses ke daily tools hain. In sab ke n8n nodes ready hain; bas credential connect karo aur operation chuno.

- Sheets = digital register
- Gmail labels = almirah ke folders
- Telegram bot = 24×7 receptionist

📖 Ek kahani se samjho: Clinic ka smart receptionist bot

1 📱 **Patient ne Telegram kiya "/book"**
= Telegram Trigger

2 📅 **Free slots check**
= Google Calendar

3 🔘 **Buttons mein slots bheje**
= Telegram (inline keyboard)

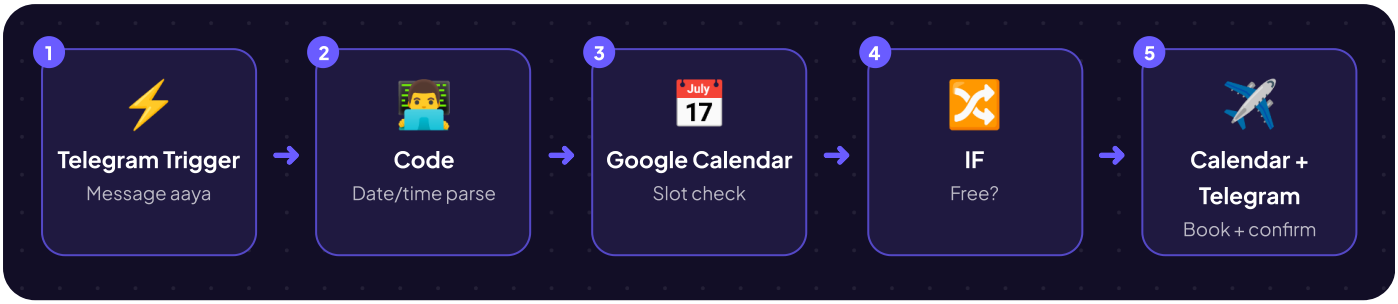
4 ✅ **Slot book**
= Calendar create event

5 ⏰ **1 ghanta pehle reminder**
= Wait + Telegram

🪄 Ek line mein: “Socho tumhare paas ek robot hai jo tumhari diary (Calendar), register (Sheet), aur phone (Telegram) sab sambhaal leta hai.”

🗺️ Aaj ka workflow: Telegram Appointment Booking Bot

Data left se right chalta hai. Har dabba ek n8n node hai.



💰 Isse kitna kama sakte ho?

Doctors, salons, consultants (appointment bot) — aam taur par **₹10,000 – ₹30,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Telegram Trigger**
User likhta hai "/book"

2 📅 **Google Calendar**
Agle 3 din ke free slots check

3 ✈️ **Telegram**
Slots ko inline buttons mein bhejo

4 📅 **Google Calendar**
Selected slot par event create

5 📊 **Google Sheets**
Booking log

6 ⏳ **Wait**
1 ghanta pehle tak ruko → reminder bhejo

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Telegram bot token (BotFather se)**
- [ ] Connect: **Google Calendar (Google OAuth)**
- [ ] @BotFather se naya bot banao aur token Telegram credential mein daalo
- [ ] Workflow Active karo (Telegram trigger ko public URL chahiye)

✅ Test kaise karein

1. Bot ko "hi" bhejo → help message aana chahiye
2. "book <kal ki date> 16:00" bhejo → Calendar check karo
3. Same slot dobara book karo → busy reply

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)

Phone uthana, slots dekhna	5 min
Diary mein likhna	2 min
Reminder call	3 min
	10 min

⚡ Ab (n8n)

Bot 24×7 booking + reminder

0 min

⏱ Har appointment par 10 min, aur no-shows kam.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
{{ $json.message.chat.id }}  // Telegram reply ke liye chat ID
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "message": {
    "chat": {
      "id": 55501
    },
    "text": "/book"
  }
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "booked": "2026-12-03T17:00:00+05:30",
  "reminder": "scheduled"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Customer Telegram par likhta hai "book 2026-12-05 16:00". Bot Google Calendar mein slot check karta hai — khaali ho to 30 min ki meeting book karke confirm karta hai, warna busy bata deta hai.

Flow: Telegram Trigger (Message aaya) → Code (Date/time parse) → Google Calendar (Slot check) → IF (Free?) → Calendar + Telegram (Book + confirm)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Telegram Appointment Booking Bot" (nodes: Telegram Message, Parse Request, Format OK?, Check Slot, Slot Free?, Book Meeting, Confirm, Busy Reply, Help Reply).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **QR scan karo** ya link kholo — workflow ki JSON file khulegi.
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
9 nodes · ready to import
adifyskills.com/workflows/day-09-telegram-appointment-bot.json

🌐 **Lesson page**
Copy buttons, quiz aur video links
adifyskills.com/day/9

Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Telegram bot token (BotFather se)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Calendar (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.

⚠️ Yeh galtiyan sabse common hain

- ! Google Sheet ka "Range" galat — header row ignore ho jaata hai
- ! WhatsApp par 24 ghante ke baad free-text bhejna (template chahiye)
- ! Calendar timezone set na karna

⭐ Pro tips — time bachane wali tricks

- ★ Sheets mein "Append or Update" se duplicates se bacho
- ★ Gmail filters (from:, has:attachment) trigger mein lagao
- ★ Telegram mein Markdown/HTML parse mode se formatting

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Telegram appointment bot jo slots check kare, Calendar mein book kare aur reminder bheje.

📝 Homework

Gmail inbox ke invoices ko automatically Drive folder (month-wise) mein save karo.

- ☐ Homework ho gaya

BEGINNER
Gmail attachment → Drive ☐

INTERMEDIATE
Telegram se Sheet mein expense log ☐

ADVANCED
WhatsApp template + button reply flow ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Booking confirm hone ke baad meeting se 1 ghanta pehle reminder bhejo.

💡 Hint 1
Ek node hai jo kisi specific time tak ruk sakta hai.

💡 Hint 2
Wait node → Resume: "At Specified Time" → meeting start se 1 ghanta pehle ka time.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

🎯 2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Telegram bot token kahan se?

- (A) @BotFather
- (B) n8n
- (C) Meta
- (D) Google

Q2. Row exist kare to update, warna add?

- (A) Append
- (B) Append or Update (upsert)
- (C) Delete
- (D) Clear

Q3. WhatsApp par 24 hr baad message?

(A) Email
(B) Free text
(C) Approved template
(D) Nahi ho sakta

Q4. Gmail attachments ka data?

(A) headers
(B) labels
(C) json
(D) binary

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

API
Do apps ke baat karne ka tareeka. Ek app sawaal bhejta hai, doosra jawab (data) deta hai.

Token
AI ke liye text ka chhota tukda. AI ka bill tokens se banta hai.

✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 10

Databases: Postgres, Supabase, Airtable & Notion

"Sheets se aage — real databases ke saath scale karo."

☐ **Day 9 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → A
@BotFather

Q2 → B
Append or Update (upsert)

Q3 → C
Approved template

Q4 → D
binary

✅ Challenge ka solution

Confirm ke baad Wait (At Specified Time): {{ DateTime.fromISO($('Parse Request').item.json.start).minus({ hours: 1 }).toISO() }} → Telegram reminder.

📚 Official docs

- Google Sheets node ↗
- Gmail node ↗
- Telegram node ↗
- WhatsApp credentials ↗

▶ Videos

- n8n telegram bot tutorial ↗
- n8n whatsapp cloud api ↗
- n8n google sheets tutorial ↗

💬 Doubt poochho

WhatsApp par seedha message

`wa.me/919000000000?text=Day%209%20(Google%20Workspace%20%26%20Messaging%20Automation)%20mein%20doubt%20hai%3A%20`

🌐 Day 9 lesson page

Quiz, copy buttons, videos

`adifyskills.com/day/9`